

Notes: Manso (JF, 2011)

Motivating Innovation

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1 One-sentence summary

This note summarizes Manso (2011), which embeds a two-armed bandit learning problem into a dynamic principal-agent model to show that innovation incentives require tolerance or reward for early failure, rewards for long-run success, commitment to long-term contracts, job security, and timely feedback, rather than standard short-run pay-for-performance.

2 Big picture

2.1 Research question

The paper asks a very precise organizational-finance question:

How should a principal design incentives when the desired action is not merely effort, but experimentation with an uncertain and failure-prone new method?

The central tension is that innovation is not simply “more effort.” Innovation requires **exploration**: trying a new method that is initially less likely to succeed but may reveal information and generate high future payoffs. A standard pay-for-performance contract punishes failure and therefore may push the agent toward **exploitation**: repeating a conventional, safer method.

2.2 Main answer

The optimal contract that motivates innovation has three distinctive features:

- **Tolerance for early failure**: early failure is not necessarily evidence of shirking; it may be evidence that the agent actually experimented.
- **Reward for long-term success**: successful experimentation creates information that pays off later, so long-horizon rewards are essential.
- **Path-dependent compensation**: the same total number of successes can be rewarded differently depending on the timing of success and failure.

In the model, an agent who fails first and succeeds later can optimally be paid more than an agent who succeeds first and fails later. Under radical exploration, an agent who fails twice may even receive more than an agent with short-lived success.

2.3 Core economic intuition

A conventional project has a higher initial probability of success. A new project has a lower initial success probability but generates information. If the agent is punished for early failure, the agent will avoid the new method even when experimentation is valuable for the principal. Therefore, the principal must make early failure less costly to the agent and must reward the long-run payoff from learning.

3 Incremental contribution of the paper

3.1 Contribution relative to standard moral hazard models

Standard principal-agent models usually ask how to induce effort and prevent shirking. Their optimal contracts resemble pay-for-performance: reward success and punish failure. Manso changes the object being motivated. The principal wants not only effort, but **experimentation**. Because experimentation is informative but failure-prone, the optimal incentive scheme has the opposite shape from standard pay-for-performance in some states.

3.2 Contribution relative to multitask models

Multitask models such as Holmstrom and Milgrom (1991) show that strong incentives on one task can distort attention away from other tasks. Manso's model is related, but richer because the choice between exploration and exploitation is dynamic. The payoff from exploration depends on what is learned after the first outcome. This generates three predictions that a static multitask model cannot capture as directly:

- compensation should depend on the *path* of performance;
- long-term commitment is essential for exploration;
- feedback is valuable when the agent must condition second-period actions on first-period information.

3.3 Contribution to corporate governance and finance

The paper gives an efficiency-based rationale for practices that are often criticized in corporate governance:

- stock options with long vesting periods;
- option repricing after poor performance;
- golden parachutes;
- managerial entrenchment;
- debtor-friendly bankruptcy rules.

These practices may protect managers or entrepreneurs after failure. In standard effort models, this protection weakens discipline. In Manso's innovation model, this protection can be exactly what is needed to make agents willing to experiment.

4 Model primitives and notation

4.1 Timing and outcomes

There are two periods. In each period, the outcome is either success S or failure F . The agent chooses one of three actions:

- 0: shirking, with success probability p_0 ;
- 1: conventional method, with known success probability p_1 ;
- 2: new method, with unknown success probability p_2 .

The probability of success under the conventional method is known:

$$p_1 = \mathbb{E}[p_1] = \mathbb{E}[p_1 \mid S, 1] = \mathbb{E}[p_1 \mid F, 1].$$

The probability of success under the new method is unknown and is learned through experimentation:

$$\mathbb{E}[p_2 \mid F, 2] < \mathbb{E}[p_2] < \mathbb{E}[p_2 \mid S, 2].$$

The new method is initially less likely to succeed than the conventional method, but success with the new method makes it look superior:

$$\mathbb{E}[p_2] < p_1 < \mathbb{E}[p_2 \mid S, 2].$$

This is the formal definition of an exploratory technology: worse ex ante in the short run, potentially better after learning.

4.2 Action plans

An action plan is written as

$$\langle i_k^j \rangle,$$

where i is the first-period action, j is the second-period action after first-period success, and k is the second-period action after first-period failure.

The two central plans are:

- **Exploitation:** $\langle 1_1^1 \rangle$, repeat the conventional method regardless of the first-period outcome.
- **Exploration:** $\langle 2_1^2 \rangle$, try the new method first; if it succeeds, continue; if it fails, revert to the conventional method.

4.3 Private costs and moral hazard

The agent incurs private cost c_1 when using the conventional method and c_2 when using the new method. Shirking avoids these costs. Shirking is less productive:

$$p_0 < \mathbb{E}[p_i], \quad i = 1, 2.$$

The principal observes only outcomes, not actions. The contract therefore specifies wages conditional on the performance path:

$$\mathbf{w} = \{w_F, w_S, w_{SF}, w_{SS}, w_{FF}, w_{FS}\}.$$

The agent has limited liability, so all wages are nonnegative.

5 The benchmark bandit problem without agency

5.1 Expected payoff from an action plan

Let the reward from success be S and the reward from failure be F . Given an action plan $\langle i_k^j \rangle$, total expected payoff is

$$R(\langle i_k^j \rangle) = \mathbb{E}[p_i]S + (1 - \mathbb{E}[p_i])F \quad (1)$$

$$+ \mathbb{E}[p_i]\{\mathbb{E}[p_j | S, i]S + (1 - \mathbb{E}[p_j | S, i])F\} \quad (2)$$

$$+ (1 - \mathbb{E}[p_i])\{\mathbb{E}[p_k | F, i]S + (1 - \mathbb{E}[p_k | F, i])F\}. \quad (3)$$

For intuition, normalize $S = 1$ and $F = 0$. Exploitation gives

$$R(\langle 1_1^1 \rangle) = 2p_1.$$

Exploration gives

$$R(\langle 2_1^2 \rangle) = \mathbb{E}[p_2] + \mathbb{E}[p_2]\mathbb{E}[p_2 | S, 2] + (1 - \mathbb{E}[p_2])p_1.$$

5.2 When is exploration efficient?

Exploration is efficient relative to exploitation if

$$\mathbb{E}[p_2] + \mathbb{E}[p_2]\mathbb{E}[p_2 | S, 2] + (1 - \mathbb{E}[p_2])p_1 \geq 2p_1.$$

Rearranging,

$$\mathbb{E}[p_2] (1 + \mathbb{E}[p_2 | S, 2] - p_1) \geq p_1, \quad (4)$$

$$\mathbb{E}[p_2] \geq \frac{p_1}{1 + \mathbb{E}[p_2 | S, 2] - p_1}. \quad (5)$$

Equivalently,

$$\mathbb{E}[p_2] \geq p_1 - \frac{p_1(\mathbb{E}[p_2 | S, 2] - p_1)}{1 + \mathbb{E}[p_2 | S, 2] - p_1}.$$

Interpretation. The new method may be worth trying even when $\mathbb{E}[p_2] < p_1$ because first-period experimentation produces information. The right-hand-side reduction relative to p_1 is the value of experimentation.

6 Principal-agent problem

6.1 Expected payment and cost

If the agent follows $\langle i_k^j \rangle$ under contract $\mathbf{w}$, expected payments are

$$W(\mathbf{w}, \langle i_k^j \rangle) = \mathbb{E}[p_i]w_S + (1 - \mathbb{E}[p_i])w_F \quad (6)$$

$$+ \mathbb{E}[p_i]\{\mathbb{E}[p_j | S, i]w_{SS} + (1 - \mathbb{E}[p_j | S, i])w_{SF}\} \quad (7)$$

$$+ (1 - \mathbb{E}[p_i])\{\mathbb{E}[p_k | F, i]w_{FS} + (1 - \mathbb{E}[p_k | F, i])w_{FF}\}. \quad (8)$$

The agent's expected private cost is

$$C(\langle i_k^j \rangle) = c_i + \mathbb{E}[p_i]c_j + (1 - \mathbb{E}[p_i])c_k.$$

The contract implements $\langle i_k^j \rangle$ if it minimizes expected payments subject to incentive compatibility:

$$W(\mathbf{w}, \langle i_k^j \rangle) - C(\langle i_k^j \rangle) \geq W(\mathbf{w}, \langle l_n^m \rangle) - C(\langle l_n^m \rangle)$$

for all alternative plans $\langle l_n^m \rangle$.

6.2 Principal's objective

Given an implemented plan, the principal's expected profit is

$$\Pi(\langle i_k^j \rangle) = R(\langle i_k^j \rangle) - W(\mathbf{w}^*, \langle i_k^j \rangle),$$

where $\mathbf{w}^*$ is the least-cost contract implementing the plan.

7 Optimal contract for exploitation

7.1 Key parameters

Define

$$\alpha_1 = \frac{c_1}{p_1 - p_0}.$$

This is the standard incentive payment needed to induce the conventional action rather than shirking: it divides the cost of effort by the incremental probability of success.

Also define

$$\beta_1 = \frac{(\mathbb{E}[p_2] - p_0) + \mathbb{E}[p_2](\mathbb{E}[p_2 | S, 2] - p_0)}{(p_1 - p_0) + \mathbb{E}[p_2](p_1 - p_0)}.$$

This cutoff determines whether the agent is tempted to explore instead of exploit.

7.2 Contract form

The optimal contract implementing exploitation $\langle 1_1^1 \rangle$ satisfies

$$w_F = w_{SF} = w_{FF} = 0,$$

$$w_{SS} = w_{FS} = \alpha_1,$$

$$w_S = \alpha_1 + \frac{c_1(1 + \mathbb{E}[p_2])}{p_1 - \mathbb{E}[p_2]} \left(\beta_1 - \frac{c_2}{c_1} \right)^+.$$

7.3 Economic interpretation

If c_2/c_1 is high, exploration is costly for the agent. The only important deviation is shirking, so the contract resembles repeated pay-for-performance. If c_2/c_1 is low, the agent may want to explore. The principal then pays an extra first-period success reward w_S to push the agent toward the safer conventional method.

The exploitation contract is therefore short-run-performance oriented. Success is rewarded; failure is not protected. This is the opposite of what is needed to motivate experimentation.

8 Optimal contract for exploration

8.1 Moderate versus radical exploration

Exploration is **radical** if first-period failure is especially diagnostic of exploration relative to exploitation:

$$\frac{1 - \mathbb{E}[p_2]}{1 - p_1} \geq \frac{\mathbb{E}[p_2]\mathbb{E}[p_2 \mid S, 2]}{p_1^2}.$$

It is **moderate** otherwise.

The left-hand side is the likelihood ratio of early failure under exploration versus exploitation. The right-hand side is the likelihood ratio of two consecutive successes under exploration versus exploitation. Radical exploration means early failure is a strong signal that the agent was trying the new method.

8.2 General form of the exploration contract

For exploration $\langle 2_1^2 \rangle$, the optimal contract satisfies

$$w_S = w_{SF} = w_{FF} = 0,$$

$$w_{FS} = \alpha_1.$$

There is no reward for short-run success alone. The wage $w_{FS} = \alpha_1$ is paid after failure then success to induce effort in the second period after failure, when the agent has reverted to the conventional method.

The remaining payments are w_{SS} , which rewards long-run success, and possibly w_F , which rewards or protects early failure.

8.3 Why the contract rewards long-term success but not early success

Rewarding first-period success encourages the conventional method because $p_1 > \mathbb{E}[p_2]$. Early success is therefore not a clean signal of exploration. By contrast, two consecutive successes after using the new method are more informative because first-period success increases the posterior expected success probability of the new method.

Thus, delayed compensation is valuable because the second-period outcome provides extra information about the first-period action.

8.4 Why the contract may reward early failure

The exploration incentive constraint contains w_F with a positive sign. Early failure is more likely under the new method than under the conventional method, so paying after early failure helps separate exploration from exploitation. This is not risk insurance: the agent is risk neutral. It is an incentive device that makes the risky exploratory action privately attractive.

8.5 Path dependence

The exploration contract implies:

$$\text{SF compensation} = w_S + w_{SF} = 0,$$

$$\text{FS compensation} = w_F + w_{FS} \geq \alpha_1,$$

$$\text{SS compensation} = w_S + w_{SS} = w_{SS},$$

$$\text{FF compensation} = w_F + w_{FF} = w_F.$$

Therefore, an agent with performance path FS can earn more than an agent with path SF , even though both have one success and one failure. If $w_F > 0$, even FF can be paid more than SF . The contract rewards the *information pattern* associated with exploration rather than the raw number of successes.

9 Commitment and short-term contracting

9.1 Exploitation does not require commitment

The exploitation contract can be implemented through a sequence of short-term contracts. This is because exploitation does not require the principal to reward long-run information production. The same conventional action is repeated, so the principal can provide period-by-period incentives.

9.2 Exploration requires commitment

Exploration generally cannot be replicated with short-term contracts. If the principal cannot commit, she cannot credibly promise future rewards for early experimentation. The agent anticipates this and may choose the safer conventional method.

A sequence of short-term contracts can sometimes implement exploration when the new method is sufficiently cheap for the agent, but it is more costly than a long-term contract. For some parameters, exploration is not implementable at all without commitment.

Economic message. Long vesting periods, career concerns based on long-run outcomes, and other commitment devices are not merely compensation details. They are central to innovation incentives.

10 Termination and job security

10.1 Termination under exploitation

Under exploitation, termination after failure is useful because it deters shirking. Since shirking increases failure probability, the threat of dismissal gives the agent stronger incentives to use the conventional method. Therefore, excessive termination can be optimal when the goal is exploitation.

10.2 Termination under exploration

Under exploration, termination has two opposite effects:

- It deters shirking because shirking increases failure probability.
- It discourages exploration because experimentation also increases early failure probability relative to the conventional method.

Therefore, termination can either help or hurt innovation incentives.

When the key deviation is shirking, termination helps. When the key deviation is exploitation, termination hurts because it pushes the agent toward the safer conventional method. In the latter case, job security or excessive continuation may be optimal.

10.3 Corporate governance interpretation

Managerial entrenchment can protect a manager from dismissal after early failure. Standard governance arguments often treat entrenchment as bad because it weakens discipline. Manso's model shows that entrenchment may support experimentation by making the manager willing to try risky new strategies.

11 Feedback

11.1 No feedback for exploitation

If the principal wants the agent to repeat the conventional method, interim feedback is not valuable. The desired second-period action does not depend on the first-period outcome. Moreover, feedback may create additional incentive problems by changing the agent's continuation behavior.

11.2 Feedback is essential for exploration

Exploration requires contingent learning:

success after new method $\Rightarrow$ continue new method,

failure after new method $\Rightarrow$ switch back to conventional method.

If the agent does not receive timely feedback, he cannot condition the second-period action on what was learned in the first period. Therefore, feedback is essential for efficient experimentation.

Economic message. A creative organization should not simply tolerate failure silently. It must also provide information that helps agents update and redirect future actions.

12 Applications and empirical implications

12.1 Executive compensation

The model rationalizes several compensation practices:

- **Long-vesting stock options:** reward long-run success rather than short-run success.
- **Option repricing:** protects the manager after early failure and restores incentives to continue working.
- **Golden parachutes:** provide compensation after failure or termination, making risky exploration less privately costly.
- **Entrenchment:** gives job security and reduces the fear of being fired for early failure.

These practices may be especially valuable in firms or industries where innovation is important.

12.2 Bankruptcy law and entrepreneurship

Debtor-friendly bankruptcy laws provide failed entrepreneurs with a fresh start. In a standard effort model, such protection weakens incentives. In an innovation model, it may encourage socially valuable exploration because failed experiments generate knowledge and because early failure is part of the innovation process.

12.3 Intrapreneurship inside large firms

Large firms often struggle to motivate employees to innovate because lower-level employees may not receive credible long-term rewards for discoveries. Corporate cultures that tolerate failure, internal venture funds, tenure-like protection, and long-horizon incentives can substitute for explicit long-term contracts.

13 Extracted figures from the paper

13.1 Figures 1–2: The optimal contract that implements exploitation and the optimal contract that implements exploration

Read Figure 1: Under exploitation, compensation resembles repeated pay-for-performance. The agent is rewarded for success, and failure receives no protection. If exploration is cheap for the agent, the principal adds an early-success premium to prevent the agent from experimenting.

Read Figure 2: Under exploration, the contract rewards long-term success and may reward early failure. The agent who succeeds early but fails later receives less than an agent who fails early and succeeds later; when exploration is radical, even repeated failure may receive more compensation than short-lived success.

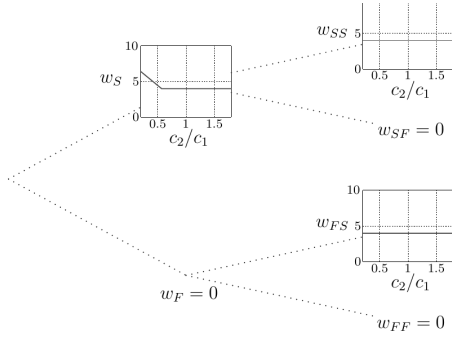


Figure 1. The optimal contract that implements exploitation under the base-case parameters. The contract resembles a repetition of standard pay-for-performance contracts. Total pay of the agent depends only on total output, except if c_2/c_1 is small, in which case the agent gets an extra reward for early success to prevent exploration.

(a) Figure 1: Exploitation contract

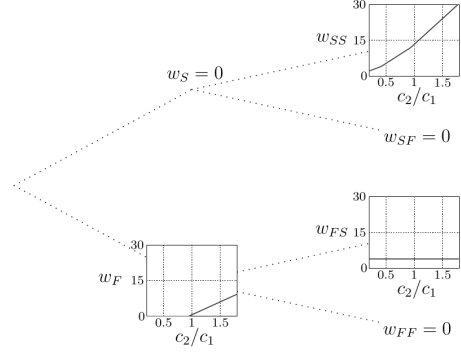


Figure 2. The optimal contract that implements exploration under the base-case parameters. Under this contract, an agent who succeeds early and fails later has lower total compensation than an agent who fails early and succeeds later or even an agent who fails twice if c_2/c_1 is high.

(b) Figure 2: Exploration contract

13.2 Figures 3–4: Commitment to long-term contracts and termination under exploitation

Read Figure 3: The cost of implementing exploration is lower with commitment to a long-term contract. Even when short-term contracts can implement exploration, they are more expensive because they cannot use delayed information as efficiently.

Read Figure 4: Termination reduces the cost of implementing exploitation. The threat of termination after early failure disciplines the agent and makes the conventional method more attractive relative to shirking or experimentation.

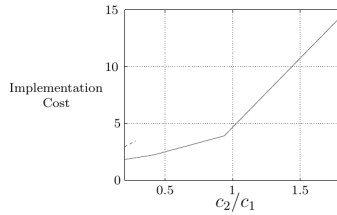


Figure 3. Cost of implementing exploration when the principal can (solid line) and cannot (dashed line) commit to a long-term contract under the base-case parameters. Even if it is feasible to motivate exploration with a sequence of short-term contracts (low c_2/c_1), it is less costly to implement exploration if the principal can commit to a long-term contract.

(a) Figure 3: Commitment and implementation cost

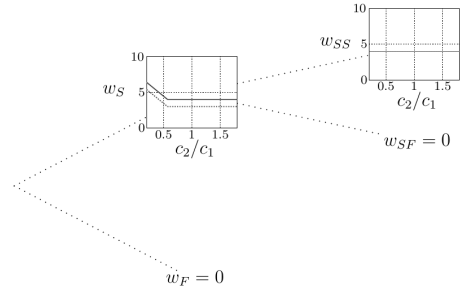


Figure 4. The optimal contracts that implement exploitation (solid line) and exploitation with termination (dashed line) under the base-case parameters. With termination, the agent loses the rents he was able to extract in the second period after a failure in the first period. The threat of termination thus induces the agent to exert more effort in the first period, and therefore the principal needs to pay less to the agent for early success in order to implement

(b) Figure 4: Exploitation with termination

13.3 Figure 5: The optimal contract that implements exploration with termination

Read Figure 5: Termination can make exploration harder to motivate when the agent is tempted to exploit. To offset the fear of being fired after early failure, the principal must provide additional protection or compensation after early failure.

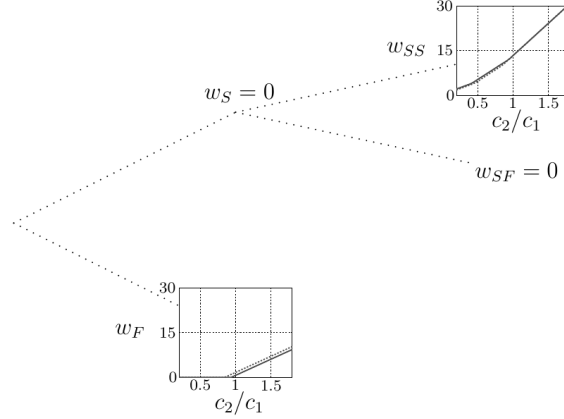


Figure 5. The optimal contract that implements exploration (solid line) and exploration with termination (dashed line) under the base-case parameters. With termination, the agent loses the rents he was able to extract in the second period after a failure in the first period. To prevent the threat of termination from drawing the agent away from exploration in the first period, the principal needs to pay an additional bonus to the agent after early failure if c_2/c_1 is high.

Figure 5: Exploration with termination

14 How to remember the paper

Dimension	Exploitation	Exploration / innovation
Economic action	Repeat known method	Try unknown method and learn
Short-run success	Strongly rewarded	Not necessarily rewarded
Early failure	Punished or receives zero	Tolerated or rewarded
Long-run success	Standard continuation reward	Central incentive instrument
Termination	Useful discipline	Ambiguous; may discourage exploration
Commitment	Not essential	Essential
Feedback	Not needed	Essential for learning
Governance analogy	Strong discipline	Job security, option repricing, golden parachutes, long vesting

Comparison between incentives for exploitation and exploration

15 Critical interpretation

15.1 Why the result is powerful

The paper explains why the same governance device can look bad in one model and good in another. A golden parachute can weaken incentives to exert standard effort, but it can strengthen incentives to try a risky new idea. The sign of a governance practice depends on the task being motivated.

15.2 Key identifying insight for empirical work

The empirical implication is not merely that generous compensation should correlate with innovation. The sharper prediction is that **path-dependent, long-horizon, failure-tolerant incentives** should correlate with innovation, especially in environments where experimentation is important and short-run failure is common.

15.3 Limitations

The model is intentionally simple: two periods, binary outcomes, one agent, and limited liability. Real innovation often involves teams, multiple stages, continuous outcomes, and strategic interactions among agents. Still, the core insight is robust: when information is produced by failure-prone experimentation, punishing early failure can destroy innovation incentives.

16 Takeaway

Manso (2011) changes the way we interpret incentives for innovation. Innovation is not motivated by simply raising pay-for-performance sensitivity. It requires a contract that distinguishes *bad failure* from *informative failure*. The optimal innovation contract therefore protects early experimentation, rewards long-run success, commits over time, provides feedback, and sometimes gives the agent job security even when short-run outcomes are poor.